FORM U-1 MANUFACTURERS' DATA REPORT FOR PRESSURE VESSELS
As Required by the Provisions of the ASME Code Rules, Section VIII, Division 1

1. Manufactured by COASTAL IRON WORKS, INC., 1133 E. Port, Corpus Christi, Tex. (Name and address of manufacturer)
2. Manufactured for Union Carbide Corp., P.O. Box 8361, So. Charleston, W.Va. 25303 (Name and address of purchaser)
3. Location of Installation Union Carbide Corp., Texas City, Texas 77590 (Name and address)
4. Type Vert. Ht. Exch. (Horiz., or vert. tank) Vessel No. 331497 (Mfgr's Serial No.) N/A (CRN) 80-2188 & 2189 (Drawing)
   312 (Nat'l Bd No.) Year Built 1981

5. The chemical and physical properties of all parts meet the requirements of material specifications of the ASME BOILER AND PRESSURE VESSEL CODE. The design, construction, and workmanship conform to ASME Rules, Section VIII, Division 1 1980 (Year) and Addenda to Winter '80 (Date) and Code Case no. N/A

   Special service per UG-120(d) N/A

   Manufacturers' Partial Data Reports properly identified and signed by Commissioned Inspectors have been furnished for the following items of the report: N/A
   (Name of part, item number, mfgr's name and identifying stamp)

Items 6-11 incl. to be completed for single wall vessels, jackets of jacketed vessels, or shells of heat exchangers

6. Shell: Material SA-106-B (Spec. No., Grade) Nominal Thickness .307 in. Corrosion Allowance 1/16 in.
   Diam. 0 ft 10-3/4 in. Length 19 ft 11-3/4 in.

CERTIFICATE OF COMPLIANCE

We certify that the statements made in this report are correct and that all details of design, material, construction, and workmanship of this vessel conform to the ASME Code for Pressure Vessels, Section VIII, Division 1.

Date 3-17-81 Signed Coastal Iron Works, Inc (Manufacturer) by E.E. True, III (Representative)

"U" Certificate of Authorization No. 3450 expires March 30 19 84

CERTIFICATE OF SHOP INSPECTION

Vessel made by Coastal Iron Works, Inc. at 1133 E. Port, Corpus Christi, Tex

I, the undersigned, holding a valid commission issued by the National Board of Boiler and Pressure Vessel Inspectors and/or the State or Province of Texas and employed by Commercial Union Ins. of Boston, Mass. have inspected the pressure vessel described in this Manufacturers' Data Report on 3/30 19 81 and state that, to the best of my knowledge and belief, the Manufacturer has constructed this pressure vessel in accordance with ASME Code, Section VIII, Division 1.

By signing this certificate neither the Inspector nor his employer makes any warranty, expressed or implied, concerning the pressure vessel described in the Manufacturers' Data Report. Furthermore, neither the Inspector nor his employer shall be liable in any manner for any personal injury or property damage or a loss of any kind arising from or connected with this inspection.

Date 3/30/81 Signed [signature] (Inspector) Commissions [illegible] (Nat'l Board, State, Province and No.)

CERTIFICATE OF COMPLIANCE FOR FIELD WORK

We certify that the statements made in this report are correct and that all details of design, material, construction, and workmanship of this vessel conform to the ASME Code for Pressure Vessels, Section VIII, Division 1.

Date _____ Signed _____ (Manufacturer) by _____ (Representative)

"U" Certificate of Authorization No. _____ expires _____, 19____

CERTIFICATE OF FIELD ASSEMBLY INSPECTION

I, the undersigned, holding a valid commission issued by the National Board of Boiler and Pressure Vessel Inspectors and/or the State or Province of _____ and employed by _____ of _____ have compared the statements in this Manufacturers' Data Report with the described pressure vessel and state that parts referred to as data items _____, not included in the certificate of shop inspection, have been inspected by me and that, to the best of my knowledge and belief, the Manufacturer has constructed and assembled this pressure vessel in accordance with ASME Code, Section VIII, Division 1.

The described vessel was inspected and subjected to a hydrostatic test of _____ psi.

By signing this certificate neither the Inspector nor his employer makes any warranty, expressed or implied, concerning the pressure vessel described in this Manufacturers' Data Report. Furthermore, neither the Inspector nor his employer shall be liable in any manner for any personal injury or property damage or a loss of any kind arising from or connected with this inspection.

Date _____ Signed _____ (Authorized Inspector) Commissions _____ (Nat'l Board, State, Province and No.)

This form (E00108) may be obtained from the Order Dept., ASME, 345 E. 47th St., New York, N.Y. 10017

FORM U-1 (BACK)

7. Seams: Longitudinal Seamless (Welded, Dbl., Sngl., Lap, Butt) R.T. N/A (Spot or Full) Efficiency 100 %
H.T. Temp N/A F Time N/A Girth Single Butt (Welded, Dbl., Sngl., Lap, Butt)
R.T. Spot (Spot, Partial, or Full) No. of Courses 1

8. Heads: (a) Material _____ (Spec. No., Grade) (b) Material _____ (Spec. No., Grade)

	Location (Top, Bottom, Ends)	Minimum Thickness	Corrosion Allowance	Crown Radius	Knuckle Radius	Elliptical Ratio
(a)						
(b)						

	Conical Apex Angle	Hemispherical Radius	Flat Diameter	Side to Pressure (Convex or Concave)
(a)				
(b)				

If removable, bolts used (describe other fastenings) _____ (Material, Spec. No., Gr., Size, No.)

9. Type of Jacket _____ Proof Test _____
10. Jacket Closure _____ (Describe as ogee & weld, bar, etc.) If bar, give dimensions _____
If bolted, describe or sketch.

11. Constructed for max. allowable working pressure 300 psi at max. temp. 300 F Min. temp. (when less than -20 F) N/A F. Hydrostatic, [redacted] test pressure 455 psi.

Items 12 and 13 to be completed for tube sections

12. Tubesheets: Stationary—Material SA-182-F304 (Spec. No. Gr.) Diam. 10.136 (Subject to pressure) in.
Nominal Thickness 1.875 in. Corrosion Allowance 0 in. Attachment Welded (Welded, Bolted)
Floating—Material N/A (Spec. No., Gr.) Diam. N/A in.
Nominal Thickness N/A in. Corrosion Allowance N/A in.
Attachment N/A

13. Tubes: Material SA-249-TP304 (Spec. No., Gr.) O.D. .625 in. Nominal Thickness 16 ~~BWG~~ gauge
Number 110 Type Straight (Straight or "U")

Items 14-17 incl. to be completed for inner chambers of jacketed vessels or channels of heat exchangers

14. Shell: Material SA-312-TP304 (Spec. No., Gr.) Nominal Thickness .365 in. Corrosion Allowance 0 in.
Diam. 0 ft 10-3/4 in. ~~O.D.~~ Length 1 ft 2 in.

15. Seams: Longitudinal _____ (Welded, Dbl., Sngl., Lap, Butt) R.T. N/A (Spot or Full) Efficiency 85 %
H.T. Temp N/A F Time N/A Girth Single Butt (Welded, Dbl., Sngl., Lap, Butt)
R.T. Spot (Spot, Partial or Full) No. of courses 1

16. Heads: (a) Material A-403-WP304 (Spec. No., Grade) (b) Material Same (Spec. No., Grade)

	Location (Top, Bottom, Ends)	Minimum Thickness	Corrosion Allowance	Crown Radius	Knuckle Radius	Elliptical Ratio
(a)	Top	.1875"	0	N/A	N/A	2:1
(b)	Btm.	Do	0	Do	Do	Do

	Conical Apex Angle	Hemispherical Radius	Flat Diameter	Side to Pressure (Convex or Concave)
(a)	N/A	N/A	N/A	Concave
(b)	Do	Do	Do	Do

If removable, bolts used (describe other fastenings) SA-193-B7; 1" dia. (16) pcs. (Material, Spec. No., Gr., Size, No.)

17. Constructed for max. allowable working pressure 400 psi at max temp. 300 F. Min. temp. (when less than -20 F) N/A F. Hydrostatic, [redacted] test pressure 689 psi.

Items below to be completed for all vessels where applicable

18. Safety Valve Outlets: Number N/A Size N/A Location N/A

19. Nozzles:

Purpose (Inlet, Outlet, Drain)	Number	Diam. or Size	Type	Material	Nominal Thickness	Reinforcement Material	How Attached
	1	4"300#LJ	Flg.	SA-312-TP304	.237"	N/R	Welded
	1	3"	Do	Do	.216"	Do	Do
	2	2"300#FFSO		SA-106-B	.218"	Do	Do
	2	1"3000#	Cplg.	SA-105	N/A	Do	Do
	5	1"	Do	SA-182-F304	Do	Do	Do

20. Inspection Openings:
Manholes No. N/A Size N/A Location N/A
Handholes No. N/A Size N/A Location N/A
Threaded No. N/A Size N/A Location N/A

21. Supports: Skirt No (Yes or no) Lugs 2 (No.) Legs N/A (No.) Other N/A (Describe)
Attached Welded to Shell (Where and how)

22. Remarks: One (1) Heat Exchanger, Item Number BX11-E-306, designed and fabricated per the ASME Boiler & Pressure Vessel Code, Section VIII, Division I, 1980 Edition and TEMA Class B.